A Basic Direction That Is Not Feasible at a Degenerate BFS

Bertsimas & Tsitsiklis, Section 3.1, Remark (b)

Statement

If $\mathbf{x}$ is a degenerate basic feasible solution, the jth basic direction $\mathbf{d}$ is not always a feasible direction. A basic variable $x_{B(i)}$ can be zero while the corresponding component $d_{B(i)}$ of $\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_j$ is negative. Moving along $\mathbf{d}$ then violates $x_{B(i)} \geq 0$ immediately.

Problem data

Consider the polyhedron in standard form ($m = 2$, $n = 4$):

$$\begin{aligned} x_1 + x_3 &= 1, \\ x_2 + x_3 - x_4 &= 0, \\ x_1, x_2, x_3, x_4 &\geq 0, \end{aligned} \qquad \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

A degenerate basic feasible solution

Choose the basis $B(1) = 1$, $B(2) = 2$. Then

$$\mathbf{B} = [\mathbf{A}_1 \ \mathbf{A}_2] = \mathbf{I}, \qquad \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

so, with the nonbasic variables $x_3 = x_4 = 0$,

$$\mathbf{x} = (1, 0, 0, 0).$$

This point satisfies $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \geq 0$, so it is a basic feasible solution. It is *degenerate* because the basic variable $x_{B(2)} = x_2$ equals 0.

The basic direction for $j = 3$ is infeasible

Set $d_3 = 1$ and $d_4 = 0$. Then

$$\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_3 = -\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \qquad \mathbf{d} = (-1, -1, 1, 0).$$

Here $x_{B(2)} = x_2 = 0$ and $d_{B(2)} = d_2 = -1 < 0$, which is exactly the situation described in the statement.

Verification. We have $\mathbf{Ad} = -\mathbf{A}_1 - \mathbf{A}_2 + \mathbf{A}_3 = \mathbf{0}$, so the equality constraints are preserved along $\mathbf{d}$. For any $\theta > 0$,

$$\mathbf{x} + \theta\mathbf{d} = (1 - \theta,\ -\theta,\ \theta,\ 0),$$

and its second component is $-\theta < 0$. Therefore $\mathbf{x} + \theta\mathbf{d} \notin P$ for every $\theta > 0$, and $\mathbf{d}$ is not a feasible direction at $\mathbf{x}$. In the simplex method, this gives the step length $\theta^* = 0$.

Contrast: the basic direction for $j = 4$ is feasible

At the same point, set $d_4 = 1$ and $d_3 = 0$:

$$\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_4 = -\begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{d} = (0, 1, 0, 1).$$

For every $\theta \geq 0$,

$$\mathbf{x} + \theta\mathbf{d} = (1,\ \theta,\ 0,\ \theta) \geq 0,$$

so this direction is feasible. Hence, at a degenerate basic feasible solution, some basic directions may be feasible and others not. This is why the text says that $\mathbf{d}$ is *not always* feasible.